

Algebra 2 (Advanced): Answer Key

Part 1: Multi-Step & Rational Exponents You Try!

1. $3(x-2)^{1/3} = 18 \rightarrow (x-2)^{1/3} = 6$

$x-2 = 216 \rightarrow$ **Answer: 218**

2. $(x-2)^{1/3} = 4 \rightarrow x-2 = 64$

$x = 66 \rightarrow$ **Answer: 66**

3. $x^{2/3} = 25 \rightarrow x = \pm 25^{3/2} = \pm(\sqrt{25})^3$

$x = \pm 125 \rightarrow$ **Answer: ± 125**

4. $(x-1)^{4/3} = 16 \rightarrow x-1 = \pm 16^{3/4}$

$x-1 = \pm(\sqrt[4]{16})^3 = \pm 8$

$x = 9, -7 \rightarrow$ **Answer: 9, -7**

1. $x^2 - 1 = 2^3 \rightarrow x^2 = 9$

Answer: ± 3

2. $x + 10 = \pm 25^{3/2} = \pm 125$

$x = 115, -135 \rightarrow$ **Answer: 115, -135**

3. $x = \pm 81^{3/4} = \pm 27$

Answer: ± 27

4. $\sqrt[3]{L} = 0.1(15) = 1.5 \rightarrow L = 1.5^3$

Answer: 3.4

5. $x - 2 = 1^{12} = 1 \rightarrow$ **Answer: 3**

6. $x = 128^{8/7} = (2^7)^{8/7} = 2^8$

Answer: 256

7. $7x^2 - 42x - 1 = 3x + 27$

$7x^2 - 45x - 28 = 0 \rightarrow (7x+4)(x-7) = 0$

Answer: 7, -4/7

Part 2: Radicals on Both Sides

5. $2x - 1 = 3x + 5 \rightarrow -x = 6$

Answer: -6

6. $2^3(5p^2) = 8p + 144 \rightarrow 40p^2 - 8p - 144 = 0$

$5p^2 - p - 18 = 0 \rightarrow (5p+9)(p-2) = 0$

Answer: 2, -1.8

Part 3: Polynomial Factoring

7. $x^2 - 2x = (-1)^3 \rightarrow x^2 - 2x + 1 = 0$

$(x-1)^2 = 0 \rightarrow$ **Answer: 1**

8. $9m^2 - 8m = m^3 \rightarrow m^3 - 9m^2 + 8m = 0$

$m(m-8)(m-1) = 0 \rightarrow$ **Answer: 0, 1, 8**

9. $15x^2 - 7x - 19 = x^2 - 7x + 6 \rightarrow 14x^2 = 25$

$x^2 = \frac{25}{14} \rightarrow$ **Answer: $\pm \frac{5}{\sqrt{14}}$**

10. $x^2 - 16x - 1 = (-4)^3 \rightarrow x^2 - 16x - 1 = -64$

$x^2 - 16x + 63 = 0 \rightarrow (x-9)(x-7) = 0$

Answer: 9, 7

Part 4: Applications

11. $\sqrt[3]{L} = 50 - 100(0.4) = 50 - 40$

$\sqrt[3]{L} = 10 \rightarrow L = 10^3 = 1000$